

Knowledge Graphs & GraphRAG: Architecture, Mechanics, and Practical Applications

Topic: Comprehensive guide to Knowledge Graphs (KGs), GraphRAG (Hybrid Graph + Vector RAG), graph construction pipelines, and their practical application across active workspace projects.

□ Visual Summary & Decision Architecture

flowchart TD

```
A["Information Retrieval & Reasoning Problem"] --> B{"What is the query & data nature?"}
```

```
%% Vector RAG Branch
```

```
B -- "Unstructured prose,
Localized semantic similarity,
'Find text snippet similar to query'" --> C["Standard Vector RAG"]
C --> C1["Chunking + Dense Embeddings"]
C1 --> C2["Top-K Cosine Similarity Retrieval"]
C2 --> C3["Strengths: Fast, Cheap, Fuzzy semantic search
Weakness: Blind to multi-hop links & global themes"]
```

```
%% Pure Knowledge Graph Branch
```

```
B -- "Strict relational facts,
Deterministic graph traversal,
'Find all cases before Justice X citing Section Y'" --> D["Deterministic Knowledge Graph"]
D --> D1["Entity & Triplet Extraction (Subject-Predicate-Object)"]
D1 --> D2["Graph DB (Kùzu / Neo4j / NetworkX) + Cypher"]
D2 --> D3["Strengths: 100% Explainable, 0 Hallucination, Multi-hop joins
Weakness: Requires structured ontology, misses fuzzy prose"]
```

```
%% Hybrid GraphRAG Branch
```

```
B -- "Complex corpus synthesis,
Multi-hop reasoning + Global summarization,
'What are recurring legal arguments across all 2024 orders?'" --> E["GraphRAG (Hybrid Graph + Vector)"]
E --> E1["1. Extract Entities & Relations via LLM"]
E1 --> E2["2. Hierarchical Community Detection (Leiden Algorithm)"]
E2 --> E3["3. Pre-generate Community Summaries"]
E3 --> E4["4. Dual Retrieval: Vector Chunks + Graph Communities"]
E4 --> E5["Comprehensive, Holistic Answer with Exact Entity Provenance"]
```

1. What is a Knowledge Graph?

A **Knowledge Graph (KG)** is a structured representation of real-world knowledge where **entities** (nodes) and their **relationships** (edges) are modeled as a network with explicit semantic meaning.

1.1 The Core Primitive: The Semantic Triplet

Every fact in a knowledge graph is fundamentally expressed as a **Subject-Predicate-Object (SPO)** triplet:

$$\text{\textit{(Subject)}} \rightarrow \text{\textit{Predicate / Relationship}} \text{\textit{(Object)}}$$

```
graph LR
  subgraph Legal_Domain_Example [Legal Domain Example]
    Case["Case: OA 83/2025"]
    Bench["Northern Bench, New Delhi"]
    Judge["Justice Prakash Shrivastava"]
    Act["NGT Act 2010 (Sec 14)"]
    Counsel["Adv. Sanjay Upadhyay"]
    Party["Delhi Pollution Control Committee"]

    Case -->|HEARD_AT| Bench
    Bench -->|PRESIDED_BY| Judge
    Case -->|INVOKES_STATUTE| Act
    Counsel -->|REPRESENTS| Party
    Case -->|PARTY_TO_CASE| Party
  end
```

1.2 Property Graphs (LPG) vs. RDF Ontologies

In modern AI and software engineering, two primary graph data models exist:

Dimension	Labeled Property Graph (LPG)	RDF / Semantic Web (Resource Description Framework)
Philosophy	Pragmatic, developer-friendly, performance-oriented	Academic, formal ontology, globally unique URIs
Node/Edge Attributes	Nodes and Edges can have arbitrary Key-Value properties	Everything is an atomic URI or Literal triplet
Primary Tools	Kùzu (embedded), Neo4j, Memgraph, NetworkX	Apache Jena, Ontotext GraphDB, Protégé, Stardog
Query Language	Cypher / openCypher, GQL	SPARQL

Dimension	Labeled Property Graph (LPG)	RDF / Semantic Web (Resource Description Framework)
Standard Format	JSON-LD, GraphML, Parquet/Arrow tables	Turtle (<code>.ttl</code>), N-Triples, RDF/XML
Best Used For	Fast application backends, LLM agents, GraphRAG	Formal taxonomies, medical ontologies, linked open data

[!NOTE] **Workspace Recommendation:** For Python-based agentic and LLM applications, the **Labeled Property Graph (LPG)** model using embedded engines like **Kùzu** (zero-overhead, embedded columnar graph database like DuckDB for graphs) or **NetworkX** (in-memory) is preferred over heavy RDF triplestores.

2. Why Pure Vector RAG Fails (And How Graphs Fix It)

While dense vector embeddings (like OpenAI `text-embedding-3`, BGE, or MiniLM) excel at semantic similarity, they suffer from critical architectural blind spots:

2.1 The 3 Failure Modes of Vector RAG

```
graph TD
    subgraph Vector_RAG_Limitations [Vector RAG Limitations]
        F1["1. Multi-Hop Reasoning Failure  
Vector search retrieves top-k isolated chunks.  
Cannot traverse A -> B -> C if B is in a separate document."]
        F2["2. Global Summarization Blindness  
'What are the main themes across all 500 cases?'  
Vector search cannot embed a corpus-level summary."]
        F3["3. Entity Disambiguation & Aggregation  
Cannot count, aggregate, or filter by strict properties  
(e.g., 'List all orders issued by Judge X with fine > ₹10 Lakhs')"]
    end
```

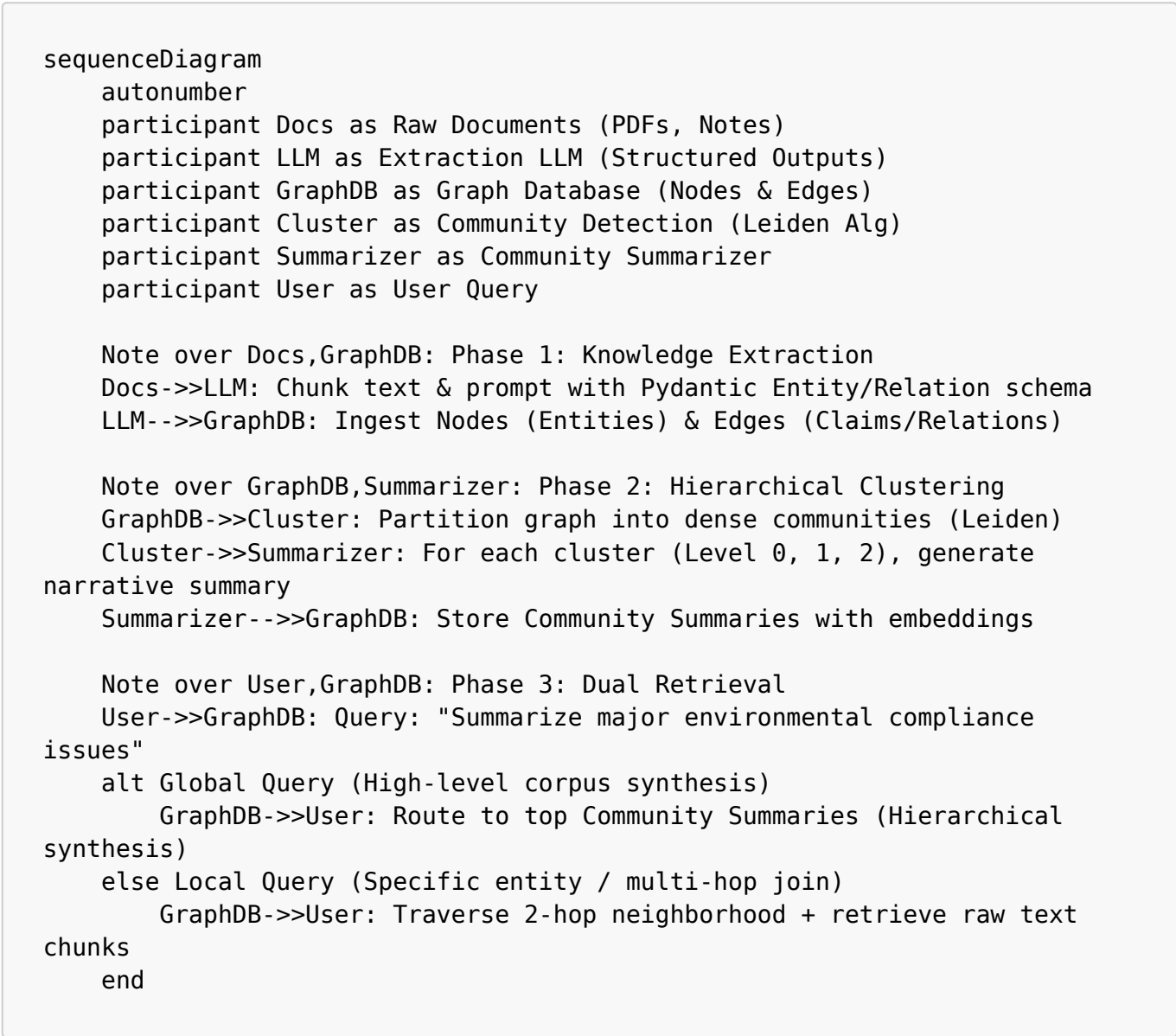
2.2 Vector RAG vs. Graph vs. GraphRAG Comparison

Capability	Vector RAG	Pure Knowledge Graph	GraphRAG (Hybrid)
Fuzzy Semantic Search	✅ Excellent	❌ Poor (Keyword / Exact match only)	✅ Excellent (Vector chunks)
Multi-Hop Traversal	❌ Blind	✅ Deterministic & Fast	✅ Graph-guided context retrieval
Global Dataset Summaries	❌ Fails (Context window overflow)	❌ Complex query writing	✅ Hierarchical Community Summaries
Structured Aggregation	❌ Hallucinates counts / filters	✅ Exact Cypher/SQL aggregations	✅ Combines structured stats + text

Capability	Vector RAG	Pure Knowledge Graph	GraphRAG (Hybrid)
Provenance & Explainability	❑ Chunk references only	❑ Explicit edge lineage	❑ Dual grounding (Edges + Chunks)
Ingestion Cost & Overhead	❑ Low (Single embedding pass)	❑ Moderate (Entity extraction)	❑ Higher (LLM extraction + clustering)

3. The GraphRAG Pipeline: How It Works

GraphRAG (pioneered by Microsoft Research and graph AI researchers) transforms unstructured documents into a structured, queryable knowledge network before retrieval.



3.1 Local Search vs. Global Search in GraphRAG

1. Local Search (Entity-Centric):
- Identifies candidate entities mentioned in the query.
 - Traverses the 1-hop and 2-hop graph neighborhood (neighbor entities, relationships, raw text chunk citations).

- *Example:* "What was the final ruling regarding the respondent Delhi Pollution Control Committee in OA 83/2025?"

2. Global Search (Corpus-Centric):

- Does *not* rely on single entity matching.
- Retrieves pre-computed **Community Summaries** across higher-level graph clusters.
- Performs a map-reduce synthesis across community reports.
- *Example:* "What are the common legal justifications for reducing environmental compensation across all cases in 2024?"

4. Application Analysis: Where Can You Use Knowledge Graphs in Your Projects?

Below is a systematic breakdown of how Knowledge Graphs and GraphRAG can be directly applied to the existing codebases in your workspace.

```
graph TD
    KG["Knowledge Graph Engine"]

    KG --> P1["LawNidhi  
(Legal Portfolio & Scraper)"]
    KG --> P2["Open-NotebookLM  
(AI Legal RAG Backend)"]
    KG --> P3["Multi-Agent Framework  
(projects/Agents)"]
    KG --> P4["Resumes & Talent Analysis  
(projects/Resumes)"]
    KG --> P5["LLM0ps Core  
(projects/llmops_core)"]

    P1 --- U1["• Case-Precedent Citation Network  
• Judge-Counsel Appearance Graph  
• Statute Violation Cross-Referencing"]
    P2 --- U2["• GraphRAG Hybrid Legal Search  
• Multi-PDF Legal Reasoning  
• Hierarchical Community Summaries"]
    P3 --- U3["• Long-Term Semantic Memory  
• Tool Dependency & SOP Graph  
• Multi-Turn State Persistence"]
    P4 --- U4["• Skill-to-Project Ontology  
• Semantic Career Trajectory Graph  
• Cross-Domain Skill Matching"]
    P5 --- U5["• Execution Lineage & Provenance  
• Prompt-Version Dependency Tracking  
• Failure Root-Cause Graph"]
```

4.1 Project 1: LawNidhi ([projects/LawNidhi](#))

□ Problem in Current Architecture

LawNidhi currently stores tabular case metadata in SQLite (`lawnidhi.db`) and downloads raw PDF orders. While SQLite handles standard relational filtering (`WHERE status = 'OPEN'`), it cannot answer:

- *"Which judicial precedents are most frequently cited by the Northern Bench when dealing with solid waste management?"*
- *"Find all instances where Senior Counsel A appeared against Respondent B before Bench C."*
- *"Show the chronological chain of orders leading up to the final judgment in Appeal 12/2023."*

□ The Legal Knowledge Graph Solution

Model LawNidhi's domain entities into a rich Property Graph:

```
graph LR
    Case["Case (0A 83/2025)"] -->|"HEARD_BY"| Bench["Bench: Northern Bench"]
    Bench -->|"MEMBER"| Judge["Judge: Hon'ble Justice P.S."]
    Case -->|"INVOKES"| Section["Section: Sec 15 (Penalties)"]
    Section -->|"PART_OF"| Act["Act: NGT Act 2010"]
    Counsel["Advocate: Mr. X"] -->|"APPEARED_FOR"| Applicant["Party: Resident Welfare Assoc"]
    Applicant -->|"FILED"| Case
    Case -->|"CITES_PRECEDENT"| PriorCase["Case: 0A 100/2018 (Vardhaman Kaushik)"]
    Order["Order: 14-03-2025.pdf"] -->|"ISSUED_IN"| Case
    Order -->|"CONTAINS_DIRECTION"| Penalty["Direction: ₹25 Lakh Fine"]
```

□ Practical Impact for LawNidhi

1. **Precedent Citation Mapping:** Automatically link orders to cited Supreme Court / High Court / NGT precedents extracted via LLM.
2. **Conflict of Interest & Bench Analytics:** Instant traversal to detect past appearances and judicial trends.
3. **Zero-Inference Structured Filtering:** Combine SQLite fast-lookup with Graph traversals for instantaneous multi-hop querying.

4.2 Project 2: Open-NotebookLM ([projects/Open-NotebookLM](#))

□ Problem in Current Architecture

Open-NotebookLM acts as the RAG backend for LawNidhi orders. Currently, it chunks PDFs into vector embeddings. When a user asks:

- *"Summarize the overarching dispute between the petitioner and the State Pollution Control Board across all 15 orders in this case."* The vector retriever fetches isolated 500-token chunks that miss the big-picture progression of hearings.

□ The GraphRAG Solution

Integrate **GraphRAG** into [workflow_engine/](#) alongside the existing vector index:

1. **Dual Indexing:** When an order PDF is uploaded:
 - Generate vector embeddings for text chunks (semantic similarity).
 - Extract legal entities, claims, and rulings into an embedded graph ([Kùzu](#) or [NetworkX](#)).
 2. **Community Summaries:** Generate cluster summaries for each legal dispute topic.
 3. **Hybrid Answer Synthesis:** Combine the specific chunk evidence with the global graph summary.
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4.3 Project 3: Multi-Agent AI Framework ([projects/Agents](#))

□ Problem in Current Architecture

Agents rely on token-limited conversation history and static SOP markdown files. Long-running tasks face context window saturation and lose long-term semantic context.

□ The Knowledge Graph Solution: Agent Semantic Memory & SOP Graphs

1. **Long-Term Memory Graph:** Store facts discovered during execution (e.g., table schemas, column data types, business rules) as graph nodes. The agent queries its memory graph instead of re-reading raw files or consuming context tokens.
 2. **Dynamic Tool & SOP Dependency Graph:** Instead of linear SOP files, model procedural steps as a Directed Acyclic Graph (DAG) of prerequisites, validations, and recovery routes.
-

4.4 Project 4: Resumes & Talent Analysis ([projects/Resumes](#))

□ Problem & Opportunity

Resumes ([master_resume.md](#), [detailed_resume.md](#), [resume.json](#)) contain rich, interconnected professional histories:

- Skills (e.g., PyTorch, Agentic AI, FastAPI)
- Projects (e.g., LawNidhi, Open-NotebookLM)
- Impacts (e.g., Latency reduction, Cost optimization)
- Domains (e.g., LegalTech, LLMOps, NLP)

□ The Skill & Experience Graph

A graph allows:

- **Transferable Skill Inference:** A query for "Distributed Systems" automatically matches candidates with "Kafka + Celery + Kubernetes" because the graph contains the parent category relationships.
 - **Tailored Resume Generation:** Automatically generate customized resume views by pruning graph subtrees relevant to a specific job description.
-

5. Technology Stack & Implementation Guide

5.1 Recommended Tooling Ecosystem

```
graph TD
  subgraph Storage & Engine Layer
    Kuzu["Kùzu DB  
(Embedded, Columnar, Lightning Fast,  
Zero Server, Cypher Support)"]
    NX["NetworkX  
(In-Memory, Pure Python,  
Graph Algorithms & Visuals)"]
    Neo4j["Neo4j / Memgraph  
(Production Server, Enterprise Scale)"]
  end

  subgraph LLM Extraction Layer
    Inst["Instructor + Pydantic  
(Guaranteed JSON Triplet Extraction)"]
    Outlines["Outlines  
(Logit-Masked FSM Extraction)"]
  end

  subgraph GraphRAG Frameworks
    MS["Microsoft GraphRAG  
(Hierarchical Leiden + Summaries)"]
    FastG["Fast-GraphRAG / LlamaIndex PropertyGraph  
(Cost-optimized, dynamic extraction)"]
  end
```

5.2 Recommended Python Stack for Your Workspace

Layer	Tool	Why It Fits Your Workspace
Embedded Graph DB	kuzu (pip install kuzu)	Runs in-process like SQLite (no Docker/server required), supports openCypher, ultra-fast columnar storage. Perfect match for LawNidhi's SQLite architecture.
Algorithm & In-Memory	networkx (pip install networkx)	Native Python graph library for clustering, shortest paths, centrality, and visualization.
Entity Extraction	pydantic + instructor	Enforces strict typed schemas for (subject, predicate, object) triplets without regex parsing.

6. Hands-On Code Blueprint: Building a Knowledge Graph with Pydantic & Kùzu

Here is a self-contained, production-ready implementation blueprint demonstrating how to extract triplets with Pydantic and store/query them in an embedded graph.

6.1 Step 1: Define the Structured Extraction Schema


```

"""schema.py: Structured extraction schema for Knowledge Graph
triplets."""
from typing import List, Literal
from pydantic import BaseModel, Field

class Entity(BaseModel):
    id: str = Field(description="Unique, normalized identifier (e.g.
'law_ngt_act_2010')")
    name: str = Field(description="Display name (e.g. 'National Green
Tribunal Act 2010')")
    entity_type: Literal[
        "CASE", "JUDGE", "BENCH", "COUNSEL", "PARTY", "STATUTE", "ORDER",
        "PENALTY"
    ] = Field(description="Category of the entity")

class Relationship(BaseModel):
    source_id: str = Field(description="ID of the source entity")
    relation_type: Literal[
        "HEARD_AT", "PRESIDED_BY", "REPRESENTS", "PARTY_TO",
        "INVOKES_STATUTE", "CITES_PRECEDENT", "ISSUED_DIRECTION"
    ]
    target_id: str = Field(description="ID of the target entity")
    weight: float = Field(default=1.0, description="Confidence or
frequency score")

class KnowledgeGraphExtraction(BaseModel):
    """Container for all extracted graph elements from a text passage."""
    entities: List[Entity] = Field(default_factory=list)
    relationships: List[Relationship] = Field(default_factory=list)

```

6.2 Step 2: Ingest & Query with Kùzu (Embedded Cypher Graph)

```

"""graph_store.py: Ingest extracted entities and run Cypher queries using
Kùzu."""
import kuzu
import shutil
import os

class LegalGraphEngine:
    def __init__(self, db_path: str = "./legal_graph_db"):
        self.db = kuzu.Database(db_path)
        self.conn = kuzu.Connection(self.db)
        self._init_schema()

    def _init_schema(self):
        """Create node and relationship tables if they don't exist."""

```

```

# Define Entity Node Table
self.conn.execute("""
    CREATE NODE TABLE IF NOT EXISTS LegalEntity(
        id STRING,
        name STRING,
        entity_type STRING,
        PRIMARY KEY (id)
    )
""")

# Define Rel Table
self.conn.execute("""
    CREATE REL TABLE IF NOT EXISTS RelatesTo(
        FROM LegalEntity TO LegalEntity,
        relation_type STRING,
        weight DOUBLE
    )
""")

def insert_graph_data(self, data: KnowledgeGraphExtraction):
    """Insert extracted nodes and edges into Kùzu."""
    # Insert Nodes
    for ent in data.entities:
        self.conn.execute(
            "MERGE (n:LegalEntity {id: $id}) "
            "ON CREATE SET n.name = $name, n.entity_type = $type",
            {"id": ent.id, "name": ent.name, "type": ent.entity_type}
        )
    # Insert Edges
    for rel in data.relationships:
        self.conn.execute(
            "MATCH (a:LegalEntity {id: $src}), (b:LegalEntity {id: "
            "$tgt}) "
            "MERGE (a)-[r:RelatesTo {relation_type: $rel}]->(b)",
            {"src": rel.source_id, "tgt": rel.target_id, "rel":
            rel.relation_type}
        )

    def find_connected_precedents(self, case_id: str):
        """2-Hop traversal: Find all precedents and statutes connected to
        a case."""
        query = """
            MATCH (c:LegalEntity {id: $case_id})-[r1:RelatesTo]->
            (target:LegalEntity)
            OPTIONAL MATCH (target)-[r2:RelatesTo]->
            (sub_target:LegalEntity)
            RETURN target.name, r1.relation_type, sub_target.name,
            r2.relation_type
        """
        response = self.conn.execute(query, {"case_id": case_id})
        return response.get_as_df()

```

7. Strategic Recommendations & Roadmap

In accordance with workspace rules regarding Scope Management, the following roadmap is suggested for planning future enhancements without immediate scope creep:

Suggested Backlog Items ([specs/backlog.md](#))

- ☐ **Backlog (LawNidhi):** [feat\(graph\)](#) - Build a local embedded graph pipeline using [kuzu](#) to map NGT judges, counsels, and cited precedents directly from parsed order PDFs.
- ☐ **Backlog (Open-NotebookLM):** [feat\(graphrag\)](#) - Implement a hybrid Graph + Vector retriever in [workflow_engine/](#) for multi-order legal summarization and cross-case thematic clustering.
- ☐ **Backlog (Agents):** [feat\(memory\)](#) - Add a lightweight [NetworkX](#) semantic memory module to persist agent learnings across multi-turn ReAct loops.
- ☐ **Backlog (Resumes):** [feat\(ontology\)](#) - Build a skill-project ontology graph to power automated, targeted resume tailoring against job descriptions.

8. Summary Checklist: When to Choose What

```
graph TD
    Q{"What is the primary retrieval task?"}

    Q -- "Keyword / ID exact lookup" --> A1["SQLite / PostgreSQL Index  
(Deterministic, O(1), zero token cost)"]
    Q -- "General text similarity / Semantic prose" --> A2["Vector RAG  
(Dense Embeddings)  
(Fast, fuzzy matching, single-doc search)"]
    Q -- "Strict relationship links / Hierarchy / Multi-hop" --> A3["Knowledge Graph (Cypher / Kùzu)  
(Explainable, multi-hop joins, precise filters)"]
    Q -- "Corpus synthesis + Multi-hop reasoning across many docs" --> A4["GraphRAG (Hybrid Graph + Vector)  
(Community summaries + dense chunk retrieval)"]
```